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MILITARY SPECIFICATION

RIG, DRILL, ROTARY AND TOP HEAD TYPE, MUD CIRCULATION

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specificaton covers a well drilling machine, consisting of a power unit, mast, hoisting assembly, rotary assembly, mud pump, and fluid circulation system with the necessary controls, all mounted on a semitrailer.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055 - Barrier Material, Waterproof, Flexible.
PPP-P-40 - Packaging and Packing of Hand Tools.
PPP-T-60 - Tape, Packaging, Waterproof.
QQ-S-781 - Strapping, Steel, and Seals.
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer Resisting.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3820

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MIL-V-173 - Varnish, Moisture-and Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-R-196 - Repair Parts, Accessories, and Kits, Mechanical; Packaging of.
MIL-L-2105 - Lubricating Oil, Gear, Multipurpose.
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal.
MIL-E-10062 - Engines: Preparation for Shipment and Storage.
MIL-H-13531 - Hose, Rubber and Hose Assembly, Rubber (Hydraulic, Flexible).
MIL-T-22085 - Tape, Adhesive, Preservation and Sealing.
MIL-P-46093 - Primer Coating, Synthetic (for Brake Drums).

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-1223- Nontactical Wheeled Vehicles; Treatment, Painting, Rustproofing, Identification Marking and Data Plate Standards.
MS 51336 - Lunette - Coupler, Drawbar, Ring.
MS 75021 - Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 28 Volt, Waterproof.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR (DoL)

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code - Section VIII.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF CALIFORNIA

California Vehicle Code.

(Application for copies should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The rig shall consist essentially of a power package, transfer case, rotary assembly, mast, fluid circulation system, draw works, and equipment for hoisting and handling drill string components.

* 3.2 First article. When specified (see 6.2), the contractor shall furnish a rig for first article inspection and approval (see 4.2.1 and 6.4).

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3.3 Standard commercial product. The drill rig shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the drill rig being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

* 3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Safety. All rotating or reciprocating parts, and all parts subject to high operating temperatures, that are of such a nature or are so located as to be or become a hazard to operating and maintenance personnel, shall be adequately guarded or insulated to the extent necessary to eliminate the hazard. All platforms, steps, and catwalks shall have antiskid surfaces. Ladders, steps, and handholds shall be provided for easy access to the control station and other work areas. The rig shall comply with OSHA regulations applicable on the date of manufacturer.

3.7 Ease of maintenance. Design of the rig shall permit ready accessibility to all parts for maintenance and repair in the field, by the use of conventional general-purpose tools associated with drill rigs. Replacement and adjustment of component parts and accessories shall be accomplished with minimum disturbance to other elements on the rig. Ease of maintenance provisions shall incorporate features insuring clearance between adjacent components to facilitate maintenance and servicing operations.

3.8 Dimensions. Overall travel width of the rig shall be not more than 96 inches. Overall shipping dimensions shall be such as to allow the rig to be towed on and off an Landing Ship Tank, with loading dimensions as indicated in figure 1.

3.9 Performance. The rig shall be capable of setting casing in 20-foot lengths and drilling holes through earth and rock formations to the dimensions specified in table I.

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TABLE I. Performance requirements.

Bit size (inches)	Depth (feet)	Casing	
		Outside diameter (inches)	Wall thickness (inch)
9-7/8	600	8.625	0.1719
7-7/8	500	6.625	0.1719
6	1,500	4.500	0.1719

3.10 Mounting. Rig shall be supported on a welded structural steel bed mounted on a semitrailer as specified herein. The bed shall be constructed essentially from structural steel channel crossmembers and headers. All exposed surface areas shall be decked over with 1/8-inch thick raised pattern steel deck plate.

* 3.10.1 Semitrailer. The steel bed shall be mounted on a 2-axle semitrailer. Frame shall be fabricated of structural steel members of suitable size and weight. The upper fifth wheel plate shall be of sufficient size to cover a lower truck fifth wheel plate of 36 inches in diameter. Forward end of the plate shall have a turned-up lip for ease of loading and for body protection. Fifth wheel shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.70(b), (e), and (h). Kingpin shall conform to SAE J700. Two vertical lift, telescopic, nonrotating, landing legs, with 2-speed gears and a handcrank shall be provided. Handcrank shall be located on the curb side of the semitrailer. With the trailer fully loaded, gross vehicle weight shall be no more than 40,000 pounds (lb), with axle loading of 17,000 lb maximum per axle. Width of the semitrailer shall not exceed 96 inches. Clearances shall preclude interference between tires and any other part of the semitrailer under operating conditions. Parked on level ground and resting level on its legs, the fully loaded, uncoupled semitrailer shall conform to the following dimensions and clearances:

a. Ground clearance - not less than 12 inches.
b. Turning clearance radius from center of kingpin to nearest portion of chassis 6 inches or more below the upper fifth wheel plate - not less than 82 inches.
c. Upper fifth wheel height from ground - 55 inches ±1 inch.
d. Swing radius from centerline of kingpin to most distant point on front end of chassis nose - not more than 56 inches.
e. Distance from kingpin to front end of chassis - not less than 18 inches.

3.10.2 Brakes. Four wheel service brakes shall be of the full air internal expanding type. Service brakes shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.40, 393.42, 393.43, and 393.45 through 393.47, and as specified herein. Braking system shall include standard breakaway features, relay emergency valve, air reservoirs, slack or automatic adjusters, piping, hose connections, gladhands, dummy couplings equipped with security chains, provisions for stowing dummy couplings, and all other

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components required to form a complete system. Air hose coupling set and location shall comply with SAE J702. The braking system shall be installed in a manner which shall provide road clearance for travel over uneven terrain and protection against damage caused by objects striking components. No part of the braking system shall extend below the bottom of wheel rims. When required, an emergency stopping system shall be furnished conforming to California Vehicle Code, section 26508. The emergency system shall also serve as parking brake and shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.41. The braking system shall meet the performance requirements of DoT Federal Motor Carrier Safety Regulation, section 393.42.

3.10.3 Wheels, rims, tires, and tubes. Semitrailer shall be equipped with single disk type wheels on each axle. Tires shall be of the low profile, high flotation type. Tire and rim sizes shall be the same for all wheels and shall conform to TRA recommendations for the type and size of tires furnished. Tires shall be of a rated capacity at least equal to the load imposed, measured at each wheel at the ground, when the semitrailer is fully loaded. Tires shall be of not less than 100 level quality and of domestic make. When tube-type tires are furnished, tubes shall be of the heavy-duty type and of proper size for the tires furnished. Tire flaps shall be provided in accordance with TRA recommendations. Unless otherwise specified (see 6.2), an inflated spare tire and wheel or rim shall be installed under the frame at curb side in an accessible location.

3.10.4 Lighting. The electrical lighting system shall be 12 volt (V) potential and shall conform to DoT Federal Motor Vehicle Safety Standards and Regulations, sections 393.14, 393.20, 393.22, 393.23, 393.25 through 393.29, 393.32, and 393.33. All lights and reflectors shall be mounted in recessed or otherwise guarded locations. Lights and reflectors shall not be mounted on rub rails or bumpers. Turn signal lamps conforming to SAE J588, class A, shall be furnished at rear of the semitrailer. All electrical wiring shall conform to SAE J555. The semitrailer shall be equipped with one 12-contact receptacle with cover and one 7-contact receptacle with cover installed at the front of the semitrailer. The 12-contact receptacle shall conform to MS 75021, part socket for jumper cable. The 12-contact receptacle shall be provided with resistance to each circuit to reduce the voltage of a tactical military design towing vehicle from 24V to 12V. The 7-contact socket shall be wired in accordance with SAE J560 to supply 12V directly to the semitrailer circuits. The 24V, 12-contact receptacle shall be connected through the necessary resistors into the 12V semitrailer circuits as follows:

A - Do not connect.
B - Connect to circuit No. 3.
C - Do not connect.
D - Connect to circuit No. 1.
E - Connect to circuits No. 2 and 6.
F - Do not connect.
H - Do not connect.
J - Connect to circuit No. 5.
K - Connect to circuit No. 7.
L - Do not connect.
M - Do not connect.
N - Do not connect.

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Circuits B and J on tactical truck-tractors are combination stop and turn indicator circuits. As the 12V turn signal lights will function both as turn signals and stop lights, the No. 4 circuit shall not be connected to the 24V, 12-contact receptacle.

3.10.5 Stabilizing jacks. Semitrailer shall be equipped with four, double acting, hydraulic, stabilizing jacks powered by the rig's hydraulic system. Extension and retraction of the jacks shall be controlled by the operator at the control console. Jacks shall have sufficient capacity to sustain the maximum loading which can be imposed by the rig. Jacks shall be provided with dual pilot check valves or counterbalance valves and mechanical locks to prevent uncontrolled retraction. Each jack shall be equipped with positive locking device to prevent accidental extension while in a travel position. Two jacks shall be located at the rear corners of the trailer. Two jacks shall be located immediately behind the upper fifth wheel plate in a position that shall conform to the dimensions and clearances as specified in 3.8.

3.10.6 Transportability. The entire rig shall be capable of being towed at speeds as great as 10 miles per hour (mph) over unimproved roads, and over reasonably hard, uneven terrain; and at speeds as great as 55 mph over improved roads as specified in 6.5.1. The semitrailer shall conform to the tracking requirements of DoT Federal Motor Carrier Safety Regulations, section 393.70(e).

3.10.7 Dolly. When specified (see 6.2), the rig shall be furished with a single axle dolly. Centerline of axle to rear of dolly shall be not more than 30 inches. Dolly shall be equipped with single wheels and low profile, high flotation tires. Rims and tires shall be the same size as those furnished on the semitrailer. Drawbar shall be hinged and when in a horizontal position shall be 36 inches above ground level. Drawbar shall be capable of being locked in eight positions in equal increments from 8 inches above ground level to 8 inches above horizontal. A lock shall be furnished to maintain the eight positions. Drawbar shall be equipped with a lunette conforming to MS 51336. Drawbar shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.70. The unladen height of the rigid-type fifth wheel shall be 55 inches +1 inch. Diameter of the fifth wheel shall be 36 inches. Fifth wheel shall be of the open-throat type coupler plate and shall be compatible with the kingpin as specified in 3.10.1. Overall width of the dolly shall be not greater than 96 inches. Dolly shall be capable of assuming an angle of 55 degrees (°) to the coupled towing vehicle without cramping or sideslipping. Without causing damage to the dolly and semitrailer, the dolly shall be capable of assuming an angle of 90° to the coupled semitrailer. Brakes, reflectors, and lights shall conform to the requirements as specified in 3.10.2 and 3.10.4. Two safety chains shall be furnished conforming to the requirements of DoT Federal Motor Carrier Safety Regulations, section 393.70(f).

3.11 Power package. The rig shall be powered by a diesel engine having horsepower, torque, and speed characteristics to meet satisfactorily all drilling performance requirements specified herein. Engine shall start within 5 minutes and be ready for full load operation within 15 minutes in any ambient temperature from +52°Celsius (C) to -29°C. The engine shall be furnished complete with the following:

a. A fuel tank of sufficient capacity for 12 hours rated load operation.

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b. 12V or 24V electrical cranking system with battery. The battery shall be 12V potential, lead acid, negative ground, and have a capacity of not less than 200 ampere-hours at the 20-hour rate. The battery shall be shipped, charged, and dry. Electrolyte shall not be furnished.
c. A battery charging alternator with a rating of not less than 60 amperes.
d. A 2-stage, dry type, intake, air cleaner with service indicator.
e. A waterproof slave receptacle, with mating plug, electrically connected to the battery charging system to facilitate emergency starting and battery charging.
f. Safety control devices of the visual or audible type for high coolant temperature and low lubricating oil pressure.
g. Hood and side doors.
h. A mechanical, liquid quantity, fuel indicator located at the tank.
i. Hour meter.
j. The control console shall be equipped with, but not necessarily limited to, the following instruments and controls:

(1) Lubricating oil pressure indicator.
(2) Cooling liquid temperature indicator.
(3) Tachometer.
(4) Battery-alternator indicator.
(5) Manual speed control.
(6) Emergency shutdown device.
(7) Normal shutdown device.

k. A weatherproof battery box painted with acid-resistant paint.

* 3.12 Power transfer case. Engine power may be transferred either by a fully enclosed, oil bath lubricated chain and sprocket or gear type transfer case, or it may be a hydrostatic drive assembly. When power is furnished by a hydrostatic drive assembly, it shall include a pump drive, gear box, and hydraulic pumps to power the rotation, pullback and pulldown of the drill string, and to power the mud pump.

* 3.13 Rotary assembly. Rotary assembly shall be either a rotary table type as detailed in 3.13.1 and 3.13.2, or it may be a hydrostatic driven top head type as detailed in 3.13.3.

3.13.1 Rotary transmission. Power to the rotary table shall be transmitted from the engine through the transfer case and a rotary transmission. The rotary transmission shall have not less than four speeds forward and one speed reverse. When the engine is operating at governed speed, the rotary transmission shall provide a maximum speed of not less than 150 revolutions per minute (rpm) to the kelly bar. By gear selection and reduced engine speed, the rotary transmission shall provide a minimum speed of not more than 15 rpm to the kelly bar. Maximum torque output at the kelly bar shall be not less than 8,000 foot pound force (ft lb). Transmission shall be fitted with easily accessible drain, and filler plugs, and a means for determining the level of lubricant, such as a dipstick. Levers for shifting gears shall be provided at the control console.

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* 3.13.2 Rotary table. Rig shall be furnished with an enclosed, oil bath lubricated, heavy-duty, rotary table having a minimum opening of not less than 7-1/2 inches. The enclosed rotary table shall be fitted with oil drain and filler plugs so located that proper servicing shall be assured. Table shall be equipped with a kelly bar drive bushing. The bevel ring gear and pinion shall be of hardened alloy steel with ground, spiral cut teeth. Pinion shaft shall be mounted on heavy-duty, antifriction type bearings. Table shall be equipped with double acting hydraulic cylinder(s) to allow horizontal movement of the table. Movement of the table from its normal operating position shall be sufficient to allow passage of 12-inch diameter tools into the hole being drilled. Table movement shall be controlled by the operator at the control console.

* 3.13.3 Top head rotation drive assembly. The top head rotation drive assembly shall be provided by a cast iron gear box, with spur reduction, mounted to the mast by means of trunnion plates. The rotational assembly shall have a minimum operational speed range of from 0-150 rpm with a maximum torque of not less than 8,000 ft lb and be infinitely variable within the range of from 0-150 rpm. Main gear box shaft of the top head rotation drive assembly shall have a minimum water course of not less than 2 inches inside diameter (id), and shall be equipped with 2-3/8-inch American Petroleum Institute Internal Flushing (API-IF) pin for connecting to the drill pipe.

* 3.14 Mast. Mast shall be constructed of steel with members jointed by using one of the arc-welding processes. The mast shall have a working capacity of not less than 40,000 lb. Mast shall be designed for disassembling into two sections for shipment. Erected height of the mast shall be sufficient to allow setting of 20-foot lengths of casing using a wire line and a casing elevator. If rotation is by rotary table the mast shall be equipped with supporting devices for securing kelly and swivel to mast for transit. Mast shall be equipped with guide rails to guide the air/mud swivel and kelly bar. Guide rails shall assure that swivel and kelly bar are retracted from the hole being drilled sufficiently to allow free passage of 12-inch casing into the hole. Two hydraulic cylinders shall be provided for erecting and lowering the mast. Plungers or piston rods shall be hard chromium-plated to resist pitting and scoring. Cylinders shall be equipped with counterbalance or throttle valves to preclude dropping of the mast, should hose or valve failure occur. A locking device shall be provided to lock the mast in its drilling position. Rig shall be fitted with a cradle to support the mast when lowered for transit. Hydraulic operating controls shall be located at the control console. If rotation is furnished by a top head style rig, the mast shall be equipped with guide rails to guide the top head, and to allow the use of 12-inch casing into the hole.

3.14.1 Racking platform. A racking platform designed to permit upright stacking of 20-foot lengths of drill pipe shall be located on the mast. The platform shall be decked with open pattern steel plate. Platform shall be provided with a guardrail. The platform shall fold to a flat position when the mast is lowered for transit. Ladder rungs shall be provided on the mast for safe access to the platform.

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* 3.14.2 Crown block. When a rotary table rig is required, sheaves shall be steel, spoke type or solid plate type, and shall be mounted on roller or ball bearings. Diameter of the kelly bar, hoisting, and sand-reel sheaves shall be not less than 10 inches. When the traveling blocks, as specified in 3.16.1, are of the two part line type, not less than two eye type anchors shall be affixed to the underside of the crown block for securing the ends of the kelly bar and main hoisting lines. Eye type anchors will not be required when three part line traveling blocks are furnished.

* 3.14.3 Jib boom. When a top head type rig is required, jib boom shall be spoke type or solid plate type and shall be mounted on roller or ball bearings.

* 3.14.4 Air/mud swivel. When a rotary table rig is required, air/mud swivel shall be of the pulldown type equipped with a yoke for anchoring the pulldown chains and wire rope. Swivel yoke shall be of sufficient width to engage guide rails, as specified in 3.14, to assure proper tracking of the swivel. Thrust capacity of the yoke shall be not less than 25,000 lb at 100 rpm. Lifting capacity of the swivel shall be not less than 40,000 lb. Thrust and lifting bearings shall be heavy-duty ball or roller type. All moving steel parts shall be heat treated. Lubrication fittings shall be provided for lubricating the moving parts. Swivel shall have a minimum water course of not less than 2 inches id. Lower end of steel tube shall be externally threaded. A kelly head adapter shall be furnished to connect lower end of the tube with upper end of the kelly bar. Gooseneck shall be equipped with a swivel type connection for connecting the upper hose as specified in 3.15.3.

* 3.14.5 Kelly bar. When a rotary table rig is required, kelly bar shall be round with a minimum outside diameter (od) of not less than 3-1/2 inches and a length of not less than 24 feet. Kelly shall have three drive flutes. The lower end of the kelly shall be a box connection equipped with a saver sub having a 2-3/8 inch API-IF pin for connecting to the drill pipe. The upper end of the kelly shall be threaded with threads that are compatible with the kelly head adapter as specified in 3.14.4.

* 3.14.6 Pulldown system. Pulldown system may consist of wire ropes, sheaves, chains, sprockets, hydraulic drive, and controls, or it may consist of chains, sprockets, hydraulic drive, cylinders and controls. Chains shall be single strand, precision roller type, and shall be of sufficient length to provide continuous feed for the specified length of at least 23 feet. Sprockets, shafts, and chains shall be heat-treated alloy steel. Shafts shall be mounted on heavy-duty bearings. The hydraulic drive shall have sufficient power to provide a maximum pulldown force of not less than 20,000 lb for rotary table type rig. Controls shall be provided to control feed, engagement, and disengagement of the drive sprockets. Feed control shall operate smoothly and shall provide variable feed and pressure rates. Disengagement of the drive sprockets shall allow free rotation of the sprockets for high speed retraction of kelly. Controls shall be located at the control console. For top head type rigs, controls shall be provided to control feed. Feed control shall operate smoothly and shall provide variable control pressure rates. Controls shall be located at the control console.

3.15 Fluid circulation system. The fluid circulation system shall consist essentially of a mud pump, suction hose, discharge line, and an air manifold.

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* 3.15.1 Mud pump. Mud pump shall be of the horizontal, duplex, double acting, piston type. Valve seats shall be hardened alloy steel having a minimum hardness of 600 Brinell. Cylinder liners and piston rods shall be hard chromium plated to a minimum thickness of not less than 0.0040 inch. Piston rod shall be provided with lubricated packing glands. The alloy steel piston body shall be equipped with readily replaceable, double lipped cups of abrasive-resistant rubber. Pump shall have a discharge capacity of not less than 150 gallons per minute (gpm), at a maximum working pressure of 310 pound-force per square inch (psi), when operating at 75 strokes per minute. Pump drive shall be enclosed in a housing and, when the drive is by sprockets and chains, shall run in oil, or the pump may be driven hydraulically. A method to drain the pump and discharge line in the mounted position shall be provided. Pullers for removing cylinder liners and valve seats, and the necessary wrenches for maintaining the pump shall be furnished.

3.15.2 Suction hose. Suction hose shall be woven-wire or braided-wire filler, helix construction type, having an id of not less than 4 inches and a length of not less than 20 feet, and shall be of good quality. One end of the hose shall be equipped with an adapter for connecting a quick-disconnect type coupler on the pump. Quick-disconnect coupler shall be readily accessible, for attachment of the suction hose, to operating personnel while standing on the ground. A strainer shall be provided on one end of the hose. A foot valve shall be provided with the strainer or a swing type check valve shall be installed in the suction line.

* 3.15.3 Discharge line. The discharge line shall consist of piping, air chamber(s), pressure gage, relief valve, control valves, hoses, and mast pipe. A 2-inch id pipe shall connect to the discharge side of the pump and run to a point near the mast where it shall be equipped with a lubricated plug valve. The discharge pipe shall be equipped with:

a. A relief valve and bypass line located as close to the mud pump as possible. Valve shall be of the adjustable type within the range 50 to 1,000 psi.
b. Air chamber(s).
c. An outlet equipped with a control valve and a 1-inch female connection threaded with National taper pipe (NPT) for connecting the mud-mixing hose as specified herein.
d. An outlet for connecting a 2-inch id discharge hose from an auxiliary mud pump.
e. A dial type pressure gage.

The mud-mixing hose shall be 1-inch id by 25-feet long, rubber, wire wrapped, 300 psi rating, of good quality, and shall be equipped with male connections. Male connections shall be 1-inch NPT. A 1/2-inch id brass nozzle, not less than 6 inches long, having a 1-1/2 inch NPT female connection bushed to 1-inch, shall be furnished. Pressure gage shall be visible to the operator at the control console and shall be calibrated from 0 to 1,000 psi. Pressure gage shall be protected to prevent any fluids and colloids from entering the gage tubes. Mast pipe shall be equipped with two, high pressure, flexible hoses. The lower hose shall connect the mast pipe to the discharge pipe as specified herein. The upper hose shall connect the mast pipe to the swivel, or top head. Swivel hose shall be of sufficient length to permit full kelly or top head travel and shall be furnished with safety chains on both ends. All threads shall be NPT.

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3.15.4 Air receiver manifold. Air receiver manifold shall connect two 600 cubic feet per minute, 100 psi, auxiliary air compressors to the rig. The receiver shall be built to conform to ASME Boiler and Pressure Vessel Code, section VIII. Manifold shall be provided with a pressure relief safety valve set at 125 psi, a moisture drain valve, four inlets, and one 2-inch id outlet for connecting the discharge line. The four inlets shall be 2-inch NPT nipples equipped with caps. The discharge line shall consist of piping, one-way valve, pressure gage, relief valve, and control valves. Discharge line shall connect the air receiver manifold to the lower mast pipe hose as specified in 3.15.3. The one-way valve shall prevent the entrance of water into the air receiver manifold from the water injection pump. Discharge line shall have two inlets equipped with control valves for connecting the discharge lines of the air line lubricator and water injection pump. The pressure gage shall be of the dial type calibrated for pressures from 0 to 500 psi, and shall be protected against the entrance of fluids and colloids. Pressure gage shall be visible to the operator at the control console. The discharge line pressure relief valve shall be of the adjustable type within the range 50 to 500 psi. All threads shall be NPT.

* 3.15.5 Air line lubricator. Air line lubricator shall be of the positive displacement type having a minimum reservoir capacity of not less than 3 gallons. Discharge rate shall be adjustable. Lubricator's discharge hose shall be connected to the air manifold discharge line as specified herein. Lubricator shall be mounted on the rig in a readily accessible location for servicing and shall be equipped with a glass sight gage to provide visual indication of oil flow.

* 3.16 Draw works. If the rig is a rotary table rig, the draw works shall consist of a main hoist, sandreel hoist, cathead, and accessories. The assemblies shall be sectionalized to permit repair or replacement of any section without the need of disassembling the hoisting equipment in its entirety. Component parts shall be interchangeable wherever possible. The draw works shall be driven by a spiral bevel ring gear and pinion or by chain and sprockets, either directly or indirectly, by the power transmission. When chain and sprockets are indirectly driven, power shall pass through a bevel gear case. The bevel gear case output shaft shall be equipped with a drive sprocket. Shafts, sprockets, and chains shall be of heat treated, alloy steel. Chains shall be of the multiple strand, precision roller type. The bevel ring gear and pinion of the two drives shall be of hardened alloy steel with ground, spiral cut teeth. Both cases shall be fitted with oil drain and filler plugs located to assure proper servicing. All shafts of the hoisting equipment shall be mounted on heavy-duty ball or roller bearings. Draw work controls shall be located at the control console. Top head drive draw works shall be manufactures standard unit.

* 3.16.1 Main hoist. If the rig is a rotary table rig, the main hoist shall be of the double drum type. Each drum shall be equipped with a clutch and brake for independent operation. Clutches shall have sufficient capacity to give each drum a bare drum line pull of not less than 15,000 lb, and a maximum speed of not less than 190 feet per minute (fpm). Brakes shall have sufficient capacity to brake and hold a minimum load of not less than 15,000 lb on the bare drum. Brake lever shall permit weight feeding and shall be provided with

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a positive locking mechanism. Each drum shall have a spooling capacity of not less than 200 feet of 5/8-inch diameter wire rope, and shall be furnished with not less than 160 feet of 5/8-inch diameter, 7 by 18, nonrotating, hemp core, improved plow steel, preformed, uncoated steel, wire rope. The free end of both ropes shall be fitted with an open socket for securing rope ends to the two eye anchors as specified in 3.14.2 or to the three part line traveling blocks. Two single sheave, steel, two part or three part line traveling blocks shall be furnished with the rig. Sheave diameter shall be not less than 10 inches. Hoisting capacity of the traveling blocks shall be equipped with a swivel type safety hook. Each hoist shall be provided with a weight indicator compatible with the traveling block furnished and shall have a capacity commensurate to the mast. Each weight indicator gage shall be located at the control console and shall be equipped with an adjustable target pointer. Main hoist for top head-type unit shall be manufactures standard unit which is equal or better than that specified for rotary table.

* 3.16.2 Sand-reel hoist. If the rig is a rotary table rig or top head, the sand-reel hoist shall have sufficient capacity to give a bare drum line pull of not less than 8,000 lb and a maximum bare drum speed of not less than 490 fpm. Brake shall have sufficient capacity to brake and hold a minimum load of not less than 8,000 lb on the bare drum. Drum shall have a spooling capacity of not less than 1,500 feet of 3/8-inch diameter wire rope, and shall be furnished with not less than 1,200 feet of 3/8-inch diameter, 6 by 19, independent wire rope core, improved plow steel, right regular lay, preformed, uncoated steel, wire rope. Free end of the rope shall be fitted with an open socket.

3.16.3 Cathead. Rotary table and top head type cathead shall have a speed range from 100 fpm to an excess of 500 fpm. The cathead shall be not less than 8 inches in diameter. Cathead shall be so mounted as to be accessible for use from the ground, and shall be fitted with a safety device to preclude the possibility of rope turns overlapping.

3.17 Dust collection system. Rig shall be furnished with a dust control system including, but not necessarily limited to, a diverter, water tank, and water injection pump. Water tank shall have a minimum capacity of not less than 100 gallons. The tank's filler cap shall be equipped with a removable strainer that shall prevent the entrance of foreign matter into the tank. Water pump shall have a minimum discharge capacity of 8 gpm at 250 psi. The pump's discharge line shall connect to the discharge line at the air receiver manifold as specified in 3.15.4.

3.18 Handling equipment. Rig shall be provided with suitable slips and built-in breakout tong system for supporting and breaking of drill pipe.

3.19 Accessories and handtools. All accessories and handtools shall be of good quality.

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3.19.1 Accessories. Accessories shall perform as specified herein. Each rig shall be furnished with the following:

Item	Quantity
a. Bailer, 4-1/2-inch od, 20 feet long, with dart valve and hoisting bail	1
b. Bit, rock, roller, 3 cone	1
(1) Medium soft formation:	
(a) 4-1/2-inch with 2-3/8-inch API regular pin	4
(b) 6-inch with 3-1/2-inch API regular pin	4
(c) 7-7/8-inch with 4-1/2-inch API regular pin	2
(d) 9-7/8-inch with 6-5/8-inch API regular pin	2
(2) Medium hard formation:	
(a) 4-1/2-inch with 2-3/8-inch API regular pin	2
(b) 6-inch with 3-1/2-inch API regular pin	2
(c) 7-7/8-inch with 4-1/2-inch API regular pin	2
(d) 9-7/8-inch with 6-5/8-inch API regular pin	2
(3) Hard formation:	
(a) 4-1/2-inch with 2-3/8-inch API regular pin	2
(b) 6-inch with 3-1/2-inch API regular pin	2
(c) 7-7/8-inch with 4-1/2-inch API regular pin	1
(d) 9-7/8-inch with 6-5/8-inch API regular pin	1
c. Bit, 3-wing, 3-blade reaming stabilizer, with 4 sets of replaceable blades for:	
(1) 4-1/2-inch with 2-3/8-inch API regular pin	1
(2) 6-inch with 3-1/2-inch API regular pin	1
(3) 7-7/8-inch with 4-1/2-inch API regular pin	1
(4) 9-7/8-inch with 6-5/8-inch API regular pin	1
d. Clamps, casing, 2 piece, with 4 bolts and nuts for:	
(1) 4.500-inch od casing	2
(2) 6.625-inch od casing	2
(3) 8.625-inch od casing	2
e. Collar, drill, 6-inch od, 2-1/4-inch bore, 10 feet long, 4-1/2-inch API regular pin up and box down with wrench flats	6
f. Chain, proofcoil, 3/8-inch by 10 feet, with direct eye slip hook each end	1
g. Hook, strainer, with 2-inch NPT pin	1
h. Elevators, universal, heavy-duty type, adjustable, 4-1/2-inch, 6-5/8-inch and 8-5/8-inch casing	2

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Item	Quantity
i. Isolator, weight, drill string, 2-3/8-inch API-IF pin up and box down, with wrench flats for use with bottom-hole tool as specified herein (when rotary table rig is specified)	1
j. Lubricant, tool joint, zinc base, 10-lb can	2
k. Mixer, mud, 1-1/2 cubic foot hopper, 1-1/2-inch NPT swivel type female coupling fluid inlet	1
l. Pick, with handle	1
m. Pipe, drill, 3-1/2-inch od, conforming to API, grade D, physical properties, 0.362-inch minimum wall thickness, with flush type weld-on tool joints. Rotary connections shall be 3-1/2-inch od, 1-3/4-inch bore, with 2-3/8-inch API-IF box and pin, having wrench flats and steel thread protectors:	
(1) 20-foot lengths	50
(2) 5-foot lengths	2
n. Plug, hoisting, ball bearing swivel bail type, with 2-3/8-inch API-IF pin, 25,000 lb capacity (when rotary table rig is specified)	2
o. Shovel, round point	2
p. Subs:	
(1) 2-3/8-inch API-IF box to:	
(a) 2-3/8-inch API regular box	2
(b) 3-1/2-inch API regular box	2
(c) 4-1/2-inch API regular box	2
(d) 6-5/8-inch API regular box	2
(e) 2-inch API regular box	1
(2) 4-1/2-inch API regular pin to 6-5/8-inch API regular box	2
q. Swage, casing, reshaping, 2-inch NPT female connection for:	
(1) 6.625-inch od casing	1
(2) 8.625-inch od casing	1
(3) 10.750-inch od casing	1
r. Tool for bits for bottom-hole percussion tool, 6-inch od minimum, with 4-1/2-inch API regular pin, 100 psi	1
s. Tools, fishing, with 2-3/8-inch API-IF box connection, for 3-1/2-inch drill pipe with 3-1/2-inch od flush tool joints:	
(1) Overshot, circulating and releasing, slip type, with slips	1
(2) Overshot, die, female	1
(3) Tap, taper	1

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Item	Quantity
t. Wrenches:	
(1) Bit breakers for bottom-hole percussion tool bits:	
(a) Button type	1
(2) Stillson	
(a) 24-inch size	2
(b) 36-inch size	2
(c) 48-inch size	2
(3) Tong, pipe, chain, adjustable for 2 to 12-inch od pipe	2
(4) Tong, tool, chain, noncrushing, friction type, for disassembling bottom-hole percussion tool	2
(5) Tong, collar, nonadjustable, opening compatible with drill collar wrench flats	2
(6) Tong, sub, control rod, bottom-hole tool	1
u. Hose and necessary hose hardware, should also be provided for connecting the compressor to drill rig when bottom-hole percussion tool is being used to drill.	

3.19.2 Handtools. Handtools shall perform as specified herein. Each rig shall be furnished with the following:

Item	Quantity
a. Brushes:	
(1) Wire, 14 inches long	6
(2) Glue, 1-3/4-inch in diameter	2
b. File, bastard, mill with handle, 12-inches long	1
c. Hammers:	
(1) Sledge, double faced, 10 lb	1
(2) Machine, 1-1/2 lb	1
d. Knife, putty, 1-1/2 inches wide	2
e. Lubrication tools:	
(1) Grease gun	1
(2) Hand pump oiler	1
(3) Pressure type oil can	1

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3.20 Hydraulic system. Hydraulic system shall be protected against the entrance of dirt or other foreign material. The system shall be complete with all operating accessories, including pumps, hydraulic cylinders, valves, reservoir, pressure relief valve(s), and a full flow 25 micron oil filter accessible for cleaning. Pumps shall supply a continous flow of hydraulic fluid, and shall have sufficient capacity to energize all hydraulically powered components. A wiper shall be provided on each hydraulic cylinder adjacent to the piston rod seal to prevent dirt and abrasives from damaging the seal or entering the cylinder. Valves shall control retracting speed of all hydraulic cylinders, or an internal flow restriction or bypass shall be provided at each end of the cylinders. Reservoir shall be baffled and shall be constructed to withstand vibrations and shocks. Reservoir shall have sufficient capacity to allow for maximum displacement of all cylinders at the same time, and to aid the heat dissipation. Return line shall terminate below the minimum oil level of the reservoir. Suction line shall be above the reservoir bottom, and shall be protected with a readily accessible strainer. Reservoir shall be equipped with a removable cover for inspection and cleaning. Provisions shall be made for filling and draining the reservoir and for checking the quantity of fluid in the reservoir. Drain plug shall be of permanent magnetic type. All high pressure hydraulic hoses and fittings shall be capable of withstanding a bursting test pressure of four times the working pressure, and a maximum operating pressure of at least two times the working pressure. High pressure hydraulic hoses shall have the physical qualitites equal to, or conforming to MIL-H-13531, type I or II, as applicable. Hydraulic oil shall be of the noncorrosive type, with characteristics to withstand a temperature range for -29°C to 163°C.

3.21 Illumination. Rig shall be adequately illuminated to permit night operations without incurring losses in drilling rates or performance. The 24V electrical system of the power package shall furnish power to the lights. Illumination equipment shall include a duplex outlet located at the power package, switches, wiring, circuit breakers, and sealed beam lights. All wiring shall be protected against wearing and weathering by rigid and flexible conduit, and insulated grommets. Switches shall be located at the control console.

3.22 Control console. Rig shall be equipped with a control console containing all the controls and instruments specified herein. Control console shall be so located as to assure the operator a clear view of the drill pipe at its point of entry into the well. All controls shall be readily accessible to the operator in his normal operating position. Instruments and gages shall be visible to the operator in his normal operating position. Each instrument, gage, and control shall be provided with permanently affixed identification markings.

3.23 Platform. Rig shall be furnished with a rear mounted driller and helper platform(s). Platform(s) shall be constructed of steel, fabricated by electric welding, and decked with open pattern steel plate. Platform(s) shall be hinged to allow raising for transit. In transit position, no portion of the platform(s) shall extend below the semitrailer's main frame. A positive locking device shall be provided to secure the platform(s) in transit position.

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* 3.24 Tool storage. Rig shall be furnished with toolboxes and built-in storage compartments to carry all tools and accessories specified herein that are 5 feet and less in length. Boxes and compartments shall be constructed of sheet steel. All doors and lids shall be equipped with continuous hinges and latches. Latches shall keep doors and lids securely fastened when subjected to vibrations. In addition to the latches, each box and compartment shall be fitted with a staple, hasp, and padlock. Not less than six keys shall be furnished for the padlocks, and any single key shall open all the padlocks. Toolboxes shall be securely fastened to the rig's bed in easily accessible locations. Built-in storage compartments shall be located under the rig's bed, but no portion of the compartments shall extend below the lowest extremity of the trailer's main frame. Compartment floors shall be decked with plywood. Plywood shall be exterior grade, 5-ply, and not less than 3/4-inch thick. Plywood shall not be spliced together to form a compartment deck.

3.25 Lifting and tiedown attachments. The rig shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the rig. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments shall be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the rig on the carrier when shipped.

3.26 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.27 Electromagnetic interference characteristics. When specified (see 6.2), the rig shall conform to the elctromagnetic characteristic requirements and test limits of SAE J551 and shall be equipped with radio noise suppressors conforming to SAE J552.

3.28 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.29 Rustproofing. The semitrailer shall be rustproofed in accordance with MIL-STD-1223.

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3.30 Identification plate. An identification plate will be furnished by the contracting officer for each rig. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each rig in a conspicuous place with nonferrous screws, rivets, or bolts not less than 1/8-inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

* 3.31 Instruction plates. The rig shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.32 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.33 Workmanship.

3.33.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.33.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.32.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

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* 3.33.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.33.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on rig when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4.7, and the packaging inspection of 4.5.

4.3 Examination. Each rig shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy

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and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests.

4.4.1 Road test. Rig shall be fully equipped with tools and accessories as specified herein that are 5 feet and less in length. Rig shall be coupled to a towing vehicle and towed not less than 20 miles at speeds specified in 3.10.6, of which 10 miles shall be on improved roads and 10 miles shall be over uneven terrain. At the completion of the test, rig shall be examined for evidence of misalinement, binding, or any other malfunction.

4.4.2 Turning and tracking tests. The fully equipped rig shall be tested for turning and tracking ability to verify compliance with the requirements of 3.10.6 and 3.10.7.

4.4.3 Brake tests. The fully equipped rig shall be tested to verify conformance to 3.10.2.

4.4.4 Drilling test. Rig shall be set up at a suitable drilling site that shall allow the rig to meet the test requirements specified in table II. Not less than 4 hours of drilling shall be done at night utilizing only the lights mounted on the rig for illumination, to determine conformance to the requirements of 3.21. A drilling log shall be maintained and shall indicate the following:

a. Formations encountered and at what depth.
b. Drilling time.
c. Shutdowns and reasons therefore, including service and maintenance.

Drilling log shall be legible and shall be submitted to the contracting officer. Mud fluid shall be utilized during the test, except as otherwise specified herein. The water injection pump and air line lubricators shall be used while drilling with the bottom-hole percussion tool. Rig shall be observed to determine conformance to the requirements of section 3.

TABLE II. Test requirements.

Bit		Depth	Total depth	Formation type	
Size (inches)	Type	(feet)	(feet)	Soft	Rock
9-7/8	3-Cone	130	330	X	X
7-7/8	3-Cone	130	360		X
6	Bottom-hole	140	500		X

4.4.5 Lifting and tiedown attachment tests. Lifting and tiedown attachments shall be tested to determine conformance to 3.25.

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4.4.6 Test for electromagnetic interference suppression. When electromagnetic interference suppression is required, the equipment shall be tested in accordance with SAE J551 to determine conformance to 3.27.

4.4.7 Operational test. Each rig shall be completely assembled, properly adjusted, lubricated, and otherwise serviced for operation. Rig shall be coupled to a towing vehicle and towed a sufficient distance over roads and at speeds specified in 3.10.6 to verify compliance with the requirements of 3.10.1, 3.10.2, 3.10.4, and 3.10.6. All controls shall be operated as many times as necessary to determine ease of operation, effectiveness, and responsiveness, and that the mechanisms actuated by the controls operate promptly, fully, and without restriction, malfunction, or excessive vibration. Gear cases, bearings, shafts, chain drives, and other moving parts shall be tactually examined for excessive heating or abnorml operation insofar as is practicable and possible. Failure to pass any phase of this test shall be cause for rejection.

* 4.5 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws, and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.3 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.4 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.4.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

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5.1.1.4.2 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall be secured in place with waterproof tape.

* 5.1.1.5 Engines. Engines, engine components, and accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, type I, method I.

5.1.1.6 Clutches.

5.1.1.6.1 Disk-type clutches. Surfaces of clutch assemblies, such as clutch pressure plates, springs, exposed surfaces of disk hubs and spline shafts, linkage, pins, arms, sleeves, throwout collars, and all metal facings within the housings, shall be sprayed or fogged with a thin film of primer conforming to TT-P-664 or MIL-P-46093. Coating of composition facings shall be held to a minimum. Unpainted surfaces of control mechanisms not enclosed shall be coated with type P-1 preservative. The clutches shall be completely disengaged.

5.1.1.6.2 Band and shoe type clutches. Clutch drum facings and clutch control mechanisms, not enclosed within housings, shall be sprayed or fogged with a thin film of primer conforming to TT-P-664 or MIL-P-46093. The clutches shall be completely disengaged.

5.1.1.6.3 Jaw-type clutches. Clutch jaws, shifter yokes, and machined surfaces of shafts shall be coated with type P-6 preservative. Clutch control mechanisms shall be coated with type P-1 preservative.

5.1.1.7 Gears.

5.1.1.7.1 Exposed gears. All unpainted surfaces of exposed gears shall be coated with type P-1 preservative or with primer conforming to TT-P-664 or MIL-P-46093.

5.1.1.7.2 Enclosed gears. Enclosed gears which operate in lubricants shall be preserved by having the housing filled to the operating level with one of the following applicable preservatives:

Operating lubricant	Preservative lubricant
Lubricating oil (SAE 10, 30, or 50)	P-10, grade as applicable
Gear lubricant (SAE 75, 80, 85, 90, or 140)	MIL-L-2105, grade as applicable
Lubricant other than above	Manufacturer's approved operating lubricant

Gear mechanism shall be operated to insure coating of all interior surfaces with preservative. Gear housings containing any of the above lubricants shall be identified with a weatherproof tag to indicate: "This housing is filled to the operating level with lubricant. Do not drain until first required lubrication change." Markings shall be applied to the tags with a waterproof material. The tags shall be attached in a conspicuous location.

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5.1.1.8 Drive Chains.

5.1.1.8.1 Exposed drive chains. Exposed drive chains shall be coated with enough type P-9 preservative to insure penetration of the preservative to the inner surface of the rollers, pins, and bushings. After the excess preservative has drained, the entire chain and the unpainted surfaces of the sprocket shall be coated with type P-1 preservative.

5.1.1.8.2 Enclosed chains. Enclosed chains and chain housings shall be preserved and tagged as specified in 5.1.1.7.2 for enclosed gears.

5.1.1.9 Brakes.

5.1.1.9.1 General. Clean all surfaces as required to remove contaminants. Coat unpainted exterior surfaces of rods, levers, cables, clevises, linkage, and associated parts with type P-1 preservative. Remove brakedrum covers and spray enclosed metallic components with primer TT-P-664 or MIL-P-46093. Spray coat brakedrum friction surfaces with a very thin coat (0.5-0.6 mil thick) of primer. Coat friction surfaces of cams, adjusting screws, and anchor pins with type P-6 or P-7 preservative. Back off all brakeshoe adjusters for maximum clearance between brakedrum and brakeshoes. Do not apply coating of any type to contact surfaces of electric brakes. Adjust brakeshoes to provide maximum clearance to drum surfaces and lined surfaces. In a conspicuous location in the operator's compartment, attach a weatherproof warning tag marked, "Brakes preserved. Do not apply brakes when vehicle is being moved. Use tow bar or similar arrangement for moving vehicle. Adjust brakes before placing equipment into operation." Markings shall be applied with waterproof material.

5.1.1.9.2 Air-actuated brake systems. The air-actuated brake systems shall be drained. Interior surfaces of the air-supply tanks shall be fogged with type P-10, grade 30 preservative. Excess preservative oil shall be drained. Threaded openings and the threads of drain plugs shall be coated with P-10. The drainplugs shall be reinstalled. Air line filters shall be drained and closed. The exhaust ports of relay emergency, quick-release, and relay valves not equipped with exhaust check valves shall be closed by inserting caps or plugs as applicable conforming to MIL-C-5501 or by sealing the ports with waterproof tape. Attach a waterproof tag to the valves and service lines indicating, "Remove (plugs, caps, or tape) from exhaust ports and valves before operating." Markings shall be applied with a waterproof material.

5.1.1.10 Hydraulic systems (except hydraulic brakes or steering systems). The hydraulic fluid supply tanks shall be filled to the operating level with hydraulic fluid required for operation. The pistons shall be retracted as far as practicable into the cylinders and secured. When the pistons cannot be fully retracted, the exposed portions of the piston rods (ramshafts) shall be coated with type P-6 or P-11 preservative and the coated surfaces wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, extending the wraps approximately 2 inches onto the ram cylinders. The wraps shall be secured in place with waterproof tape. When the pistons can be fully retracted, any remaining uncoated surfaces of the piston rods shall be coated with type P-1 preservative, with no wrapping required. The

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hydraulic control valves shall be secured in the neutral position and preserved as specified herein for piston rods. Hoses shall not be disconnected. A weatherproof tag shall be attached to the control lever indicating: "The hydraulic supply tank is filled to the operating level with fluid required for operation. Do not drain." Markings shall be applied to the tag with a waterproof material.

5.1.1.11 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved method IC-1 or IC-3. Unpainted surfaces shall be coated with primer conforming to TT-P-664 or MIL-P-46093. A weatherproof tag shall be attached in a conspicuous location indicating: "Belts have been (removed or released from tension.) (Install or tension) prior to operation." Markings shall be applied to the tags with a waterproof material.

5.1.1.12 Instruments. The dial glass of instruments not protected by a metal housing shall be covered individually or in groups with a fitted piece of plywood secured with tape conforming to PPP-T-60, type IV or MIL-T-22085, type II.

5.1.1.13 Pumps. Interior surfaces of pumps shall be coated with type P-21 preservative in a manner to insure thorough coating of all interior parts and surfaces. Excess preservative shall be drained. All openings into the pumps shall be sealed with caps or plugs conforming to MIL-C-5501 or waterproof tape conforming to MIL-T-22085, type II, or PPP-T-60, type IV.

5.1.1.14 Hose and fittings. Coat ferrous metal hose fittings with type P-1 preservative. Coil hose, to a safe minimum diameter and secure. Wrap coils with a waterproof barrier material conforming to MIL-B-121 or PPP-B-1055 and secure with tape conforming to PPP-T-60.

5.1.1.15 Wire rope. Wire rope shall be preserved method I by coating with type P-19 preservative. The preservative shall be applied to all surfaces of the wire rope. Excess preservative shall be allowed to dry (at least 4 hours). The wire rope shall be bundled or coiled and wrapped with a barrier material conforming to MIL-B-121, type I, grade A, class 2 or grade C, and secured.

5.1.1.16 Rubber tires. Tires mounted on the wheels (road tires) of the vehicle shall be inflated to 10 lb above pressure value recommended for maximum load. Tires mounted on rims (spare tires) shall be inflated to two-thirds of the specified operating pressure.

5.1.1.17 Radiator fronts and engine housings. Radiator fronts and any large openings in the engine housings shall be covered with barrier material conforming to PPP-B-1055, class E-2 or H-1. The barrier shall be secured with tape conforming to PPP-T-60, class 1. All vents, louvers, and other openings in engine housings shall be sealed with tape specified herein.

5.1.1.18 Control levers. Removable composition or rubber handgrips shall be removed and preserved in accordance with MIL-P-116, method IC-3. The preserved handgrips shall be placed in the toolbox on the rig. Unpainted exposed surfaces of controls shall be coated with type P-1 preservative.

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5.1.1.19 Service parts. The preservative application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve service parts. When specified (see 6.2), the service parts shall be preserved in accordance with level A requirements of MIL-R-196, or when parts are not specifically covered in MIL-R-196, an applicable submethod of preservation of MIL-P-116 shall be used.

5.1.1.20 Maintenance tools. Maintenance tools shall be preserved in accordance with level A preservation and packaging requirements of PPP-P-40.

5.1.1.21 Technical publications. Technical publications for each piece of equipment shall be preserved method IC-1 or IC-3.

5.1.2 Level C. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The rig shall be shipped uncrated. Masts which are removed from the rig for shipment shall either be shipped loose or the sections shall be bundled into secured lifts with flat steel strapping conforming to QQ-S-781, type I, class B, in combination with suitable wood blocking or battens as required to form nonshifting bundles. Disassembled components and accessories that are suitable for stowage in the storage compartments shall be placed therein, and the doors or lids securely fastened. Disassembled components and accessories that cannot be placed in the storage compartments, and require a container for safe and convenient handling, shall be packed in close fitting boxes conforming to PPP-B-621, class 2; or PPP-B-601, overseas type. The contents shall be cushioned, blocked and braced to prevent movement within the boxes. The service parts for each rig shall be packed separately in a container as specified for disassembled components and accessories. Where practicable, disassembled components accessories, and repair parts shall be positioned and secured to the rig by bolting or with strapping in such a manner as not to interfere with towing or lifting the rig with slings.

5.2.1.2 Level B. The complete rig shall be packed as specified for level A, except that the boxes shall be domestic type or class 1, as applicable.

5.2.1.3 Level C. The equipment shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Rig is intended for use in drilling various diameter holes to depths as shown in table I, through earth and rock formations to develop water wells.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Specifications, standards, and handbooks (see 2.1.1).
c. When a first article rig is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
d. When a spare tire is not to be furnished (see 3.10.3).
e. When a dolly is to be furnished (see 3.10.7).
f. When means for lubrication shall be in accordance with manufacturer's standard practice (see 3.26).
g. When conformance to electromagnetic interference characteristic requirements and test limits are required and the type required (see 3.27).
h. When fungus resistance is required (see 3.28).
i. Color of paint (see 3.32).
j. Level of preservation and level of packing required (see 5.1 and 5.2).
k. When service parts shall be preserved in accordance with MIL-R-196 (see 5.1.1.19).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations (FAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of FAR 52.227-7031 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required (see 3.2 and 6.2), the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete rig. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Definitions.

6.5.1 Improved road. An improved road is a smooth, hard surfaced road, such as a concrete or asphalt paved highway.

6.5.2 Unimproved road. An unimproved road is an unpaved, unstabilized road with an undulating surface having occasional chuckholes and exposed rocks.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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6.7 Title change. The title of this specification was changed from "Rig, Drill, Rotary, Mud Circulation" to "Rig, Drill, Rotary and Top Head Type, Mud Circulation."

Custodian:

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Preparing Activity:

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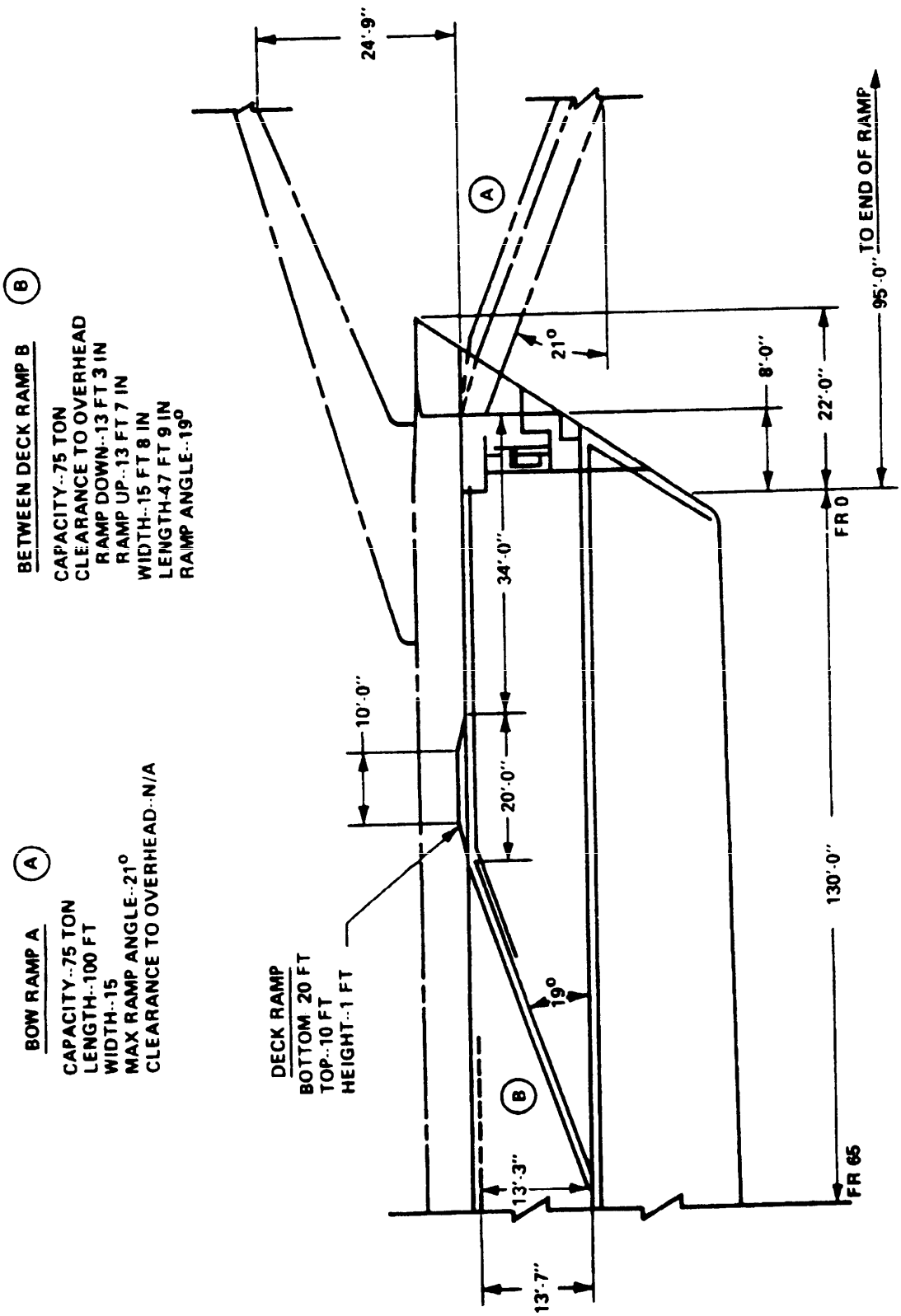


FIGURE 1. Profile, LST 1179 class bow.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-28568B(YD)	RIG, DRILL, ROTARY AND TOP HEAD TYPE, MUD CIRCULATION

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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